

CHAPTER 3

SOFTWARE DESIGN DESCRIPTION

3.1 Introduction

This chapter describes the design of the Hostel Management System for Quaid-i-Azam University. The purpose of the design phase is to convert the requirements defined in Chapter 2 into a clear technical structure.

The system design explains how users, interfaces, application modules and the database will work together. It also defines the main classes, components, workflows and user-interface screens.

The proposed system will follow a modular design. Each module will handle a specific responsibility, such as hostel applications, room allotment, complaints, payments or visitor records. This approach will make the system easier to develop, test and maintain.

3.2 System Architecture

The Hostel Management System will use a three-tier architecture.

3.2.1 Presentation Layer

The presentation layer will provide interfaces for students and staff members.

It will include:

- Login pages
- Student dashboard
- Hostel-application forms
- Application-status pages
- Complaint forms
- Room-management screens
- Payment-verification screens
- Visitor forms
- Reports dashboard

This layer will collect input from users and display results.

3.2.2 Application Layer

The application layer will process user requests.

It will handle:

- Authentication
- Role-based authorization
- Form validation
- Application processing
- Bed-allotment rules
- Complaint handling
- Payment verification
- Notice publication
- Report generation
- Audit logging

3.2.3 Data Layer

The data layer will store information in a relational database.

The main records will include:

- Students
- Users
- Roles
- Hostels
- Rooms
- Beds
- Applications
- Allotments
- Renewals
- Complaints
- Payments
- Visitors
- Notices
- Logs

3.3 System Architecture Diagram

